Early Warning, Network Stress and Cascade Amplification

Systemic Risk Engine with Early Warning Signals

Statistical and Financial Analysis (2015–2026)

Prepared for the Finance Repository and Papers Repository

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Introduction

Financial markets exhibit complex interdependencies where local shocks may propagate through network structures, generating systemic risk.

This study develops a dynamic contagion framework integrating:

- Correlation-based financial networks
- Tail risk measures (VaR, CVaR)
- Heavy-tail behavior via Hill estimator
- Threshold-based default cascades

The objective is to analyze how structural dependencies and extreme events jointly drive systemic instability. This approach extends previous work [2] by incorporating heavy-tail behavior and network-driven contagion.

Model Framework

Let $r_i(t)$ denote the log-return of asset i.

The network is defined by a weighted adjacency matrix:

$$W_{ij} = \frac{|\rho_{ij}|}{\sum_k |\rho_{ik}|}$$

where ρ_{ij} is the correlation between assets.

Edges are filtered using a threshold ρ_{min} and top-k selection.

Risk Measures

Define losses:

$$L_i = -r_i$$

Value-at-Risk:

$$VaR_i = \inf\{x : P(L_i \leq x) \geq 1 - \alpha\}$$

Conditional Value-at-Risk:

$$CVaR_i = E[L_i \mid L_i \geq VaR_i]$$

Early Warning System (EWS)

To anticipate systemic instability, we introduce an Early Warning Score (EWS) combining three key dimensions:

$$EWS_i = w_1 \cdot Stress_i + w_2 \cdot Centrality_i + w_3 \cdot TailRisk_i$$

where:

- $Stress_i = \frac{S_i}{C_i}$ measures relative shock exposure
- $Centrality_i = \sum_j W_{ij}$ captures network importance
- $TailRisk_i = \frac{1}{\alpha_i}$ reflects heavy-tail vulnerability

Weights satisfy:

$$w_1 + w_2 + w_3 = 1$$

In this model:

$$(w_1, w_2, w_3) = (0.50, 0.30, 0.20)$$

Nodes are classified as:

- **Normal:** $EWS < 0.85$
- **Warning:** $0.85 \leq EWS < 1.15$

- **Critical:** $EWS \geq 1.15$

Cascade Dynamics with Early Warning

The system evolves iteratively:

$$D_i^{(t+1)} = \mathbf{1}\left(\frac{S_i^{(t)}}{C_i} > \theta\right)$$

where:

$$S_i = \sum_j W_{ij} \cdot Shock_j \cdot D_j$$

The cascade continues until convergence:

$$D^{(t+1)} = D^{(t)}$$

The Early Warning System anticipates failures before threshold crossing, allowing identification of pre-default states.

Heavy-Tail Analysis

Tail behavior is estimated using the Hill estimator:

$$\alpha^{-1} = \frac{1}{k}\sum_{i=1}^{k} \log\left(\frac{X_i}{X_{k+1}}\right)$$

Assets with $\alpha < 3$ exhibit heavy-tail risk.

Network Interpretation

The system reveals:

- Clustering of highly correlated assets
- Shock amplification through dense subgraphs
- Asymmetry between emerging and developed markets

Brazilian assets exhibit stronger clustering, while global assets act as shock transmitters.

The inclusion of Early Warning Signals transforms the framework from a reactive model into a predictive system.

Key observations include:

- Nodes with high centrality amplify contagion risk
- Heavy-tail assets act as systemic accelerators
- Pre-default signals emerge before cascade activation

This allows early detection of systemic phase transitions.

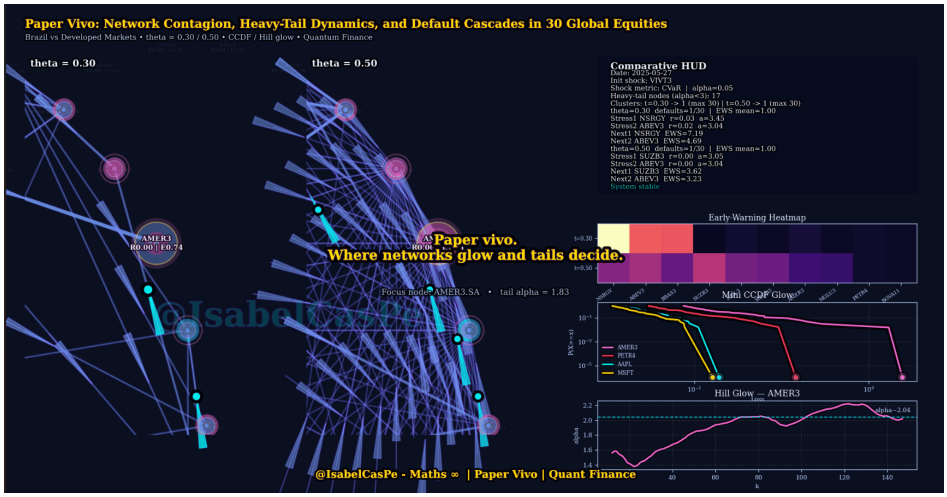

Figure 1: **Early Warning and cascade amplification in V.12** The visualization highlights pre-default dynamics through node-level Early Warning Scores (EWS), revealing how systemic stress accumulates before full cascade activation. Nodes transition from stable to critical states, reflecting the interaction between network centrality, tail risk, and shock propagation.

As shown in Figure 1, the contagion process emerges from the interaction between network connectivity, threshold effects, and tail-driven stress propagation.

Results and Visual Interpretation

The V.12 model extends the previous framework by incorporating predictive dynamics.

The animation illustrates:

- Early detection of systemic stress through EWS
- Gradual transition from stable to critical states

- Amplification driven by heavy-tail nodes

The system reveals that systemic collapse is not instantaneous but preceded by measurable structural signals.

Conclusion

The integration of Early Warning Signals significantly enhances systemic risk modeling.

The framework demonstrates that:

- Systemic crises are preceded by detectable structural stress
- Network topology and tail behavior jointly drive instability
- Early Warning Systems provide actionable insights before collapse

This establishes a predictive approach to financial contagion beyond traditional threshold-based models.

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References

[1] Castillo Pereda, A. I. (2025). *Systemic Risk and Default Cascades in Global Equity Markets: Extending the Gai-Kapadia Framework with Stochastic Simulations and Network Analysis*. arXiv:2504.01969.

[2] Castillo Pereda, A. I. (2026). *Systemic Risk and Default Cascades in Global Equity Markets: A Network and Tail-Risk Approach Based on the Gai-Kapadia Framework*. Working paper.

[3] Gai, P., & Kapadia, S. (2010). Contagion in financial networks. *Proceedings of the Royal Society A*.

[4] Glasserman, P., & Young, H. P. (2016). Contagion in financial networks. *Journal of Economic Literature*, 54(3), 779–831.

[5] Eisenberg, L., & Noe, T. (2001). Systemic risk in financial systems. *Management Science*, 47(2), 236–249.

[6] Battiston, S., Puliga, M., Kaushik, R., Tasca, P., & Caldarelli, G. (2012). DebtRank: Too central to fail? *Scientific Reports*.

[7] Barabási, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*.

[8] Mantegna, R. N. (1999). Hierarchical structure in financial markets. *The European Physical Journal B*.

[9] Forbes, K., & Rigobon, R. (2002). No contagion, only interdependence. *Journal of Finance*.